Engineering Design and Development

Overview The primary intent of this course is to provide students with the skills necessary to understand, interpret, and create engineering drawings and working sketches. Students will learn to develop 3D models and engineering drawings using Autodesk Inventor while also exploring the intersection of engineering and technical art. Through projects that emphasize visual communication, students will apply artistic principles such as multiviews, isometrics, and precision to enhance the clarity and impact of their technical designs. In addition to building spatial reasoning and technical drawing skills, students will strengthen their technical writing abilities, soft skills, and oral presentations throughout the year. The course will culminate in a 6–8 week capstone project where student teams will identify a problem, design a unique solution using Inventor, create a prototype with a 3D printer, and test the solution—integrating both engineering precision and technical artistry into their final designs. Students will also be offered the opportunity to take the Autodesk Inventor certifification exam (Advanced Statewide IBC).

Objectives

- Understanding the design intent
- Using graphics to accurately communicate information and design
- Sectioning and dimensioning multi-view drawings
- Visualizing in three dimensions
- Designing solid models.
- Creating and interpreting engineering drawings

Assessment Students will be assessed using homework assignments, quizzes, and exam as well as a final project.

Equipment	**Cost/Unit**
Inventor 3D Modeling Software	$0 for Educational Use
Computers to run Inventor (Chromebooks will NOT work)	$500 each ($0 if you already have computers). Must come with a mouse (not a tracking/touchpad)
Autodesk Inventor Certified User Exam *Advanced Statewide Industry Based Credential*	$80/student
3D Printers (Makerbot Replicator recommended) *Purchase is optional*	Reusable: Ender Crealty 5 Pro $400 each,purchase 1 per 5-7 students Consumable: filament ($200 per year)

First Semester Course Outline

Unit 1: Introduction to Design	Engineering design process, Safety, Sketching
Unit 2: Multiviews	Introduction to multi-views, Isometric view
Unit 3: Visualization	Visualization in 2D and 3D
Unit 4: Inventor – Basics	The basics of 3D modeling, sketches versus features
Unit 5: Inventor – Intermediate Part 1	Parametric modeling dimensions, Mirroring sketches and features
Unit 6: Inventor – Intermediate Part 2	Linear and Circular Patterns, Revolved features and Shells, Machined holes

Second Semester Course Outline

Unit 7: Section Views and Working Drawings (Paper and Inventor)	Defining section views, Understanding the importance of section views, Title Blocks, multi-views, sectional views, dimensioning
Unit 8: Dimensioning (Paper and Inventor)	Rules of dimensioning, Dimensioning non-standard views and holes
Unit 9: Inventor - Advanced Modeling	Lofts, Ribs, and Sweeps
Unit 10: Inventor - Assemblies and Exploded Views	Mating parts, inserting BOM, inserting balloons and part names
Unit 11: Final Project	Define, Research, Design, and Prototype Solution, 3D print and test solution, Present final solution, Project design and engineering notebooks, Writing a good problem statement, Gantt charts, Scholarly journal article review, Professionalism and ethics, Work ethic, Patents

ENGINEERING DESIGN AND DEVELOPMENT

1. Materials
   A desktop or laptop computer, access to 1-to-1 daily, and Internet. Chromebooks will not work.

Hardware/Reusable Material	Recommended Unit	Cost/Unit
3D Printer (Optional)	1 per 5-7 students	Varies*
Software (Each student needs access to a computer)		
Autodesk Inventor (current year version)	1 License per student	Free

*There are a variety of 3D printers from $500 up to $5,000. Purchase the one that best suits the needs of your school and students

2. Required software, networking access, and access to LSU servers
   - Students and teachers must have reliable access to the Louisiana Training Platform (LTP, http://louisianatrainingplatform.com)
   - Teachers will need to be able to share documents via Google drive with instructors.
   - Free software to install on each computer: Autodesk Inventor current year version. This is free for educational use and can be installed individually on each computer or via an IT roll out. More details can be found here: https://www.autodesk.com/education/edu-software/overview#INVPROSA
3. Requirements for IBC Alignment: **Autodesk Inventor**
   For schools who are pursuing the aligned IBC for this course
   a. Schools must have the appropriate testing software installed on the testing computers. For this credential the testing software is through Certiport Compass. More information can be found here: https://certiport.pearsonvue.com/Support/Install/Compass.aspx
   b. Some industry based credential providers also provide practice tests softwares such a GMetrix which may be installed. Please check with individual schools if they plan to use a practice software.
   c. Schools must also typically register as an approved testing center prior to being allowed to proctor the exams.
   d. Teachers must have earned the IBC prior to students sitting for the exam. Once they have passed the credential, they need to be set up as a proctor in the certiport system which must be done by an administrator level Certiport account.
   e. Teachers should report to the LSU Point of Contact their overall pass rates (i.e. 15/20 students passed)
4. Required teacher collaborations
   Teachers will communicate with LSU instructors via email and shared Google Drive folder. Teachers will need to share sample student work with their designated LSU Pathway Point-of-Contact.
5. Required administration of course content, pre/post test, and research instruments
   All required materials and instruments will be either posted in a Google drive or their location announced via email. Schools will need to coordinate with the Career Tech Education Manager to schedule the Inventor Industry Based Certification Exam issue through Certiport.

6. Course Work
   Teachers must present the course material in sequence or as approved by collaboration with the LSU Pathway Point-of-Contact. Teachers are expected to deliver a minimum of 80% of the course material for course fidelity and for the opportunity for IBC attainment (where applicable).
7. Other
   As this is a project-based learning class, we strongly suggest that each section of the course be limited to a *maximum* of 20 students. If the course is overloaded with students, they will not receive adequate instruction.